

# Cool Problems on Integration

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## 1. Problems

### 1.1. Based on Riemann Sum

1. Show that the Dirichlet function given by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{otherwise} \end{cases}$$

is not integrable on any interval  $[a, b]$ .

2. Show that the Riemann function given by

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational or } x = 0, \\ 1/q & \text{if } x = p/q, p \in \mathbb{Z}, q \in \mathbb{N}, \gcd(p, q) = 1. \end{cases}$$

is Riemann integrable on every interval  $[a, b]$ .

3. Let  $f : [0, 1] \rightarrow \mathbb{R}$  be defined by

$$f(x) = \begin{cases} 1 & \text{if } x = \frac{1}{n}, n \in \mathbb{N}, \\ 0 & \text{otherwise.} \end{cases}$$

Show that  $\int_0^1 f(x) dx = 0$ .

4. Find the following limits:

(a)  $\lim_{n \rightarrow \infty} \left( \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{3n} \right)$

(b)  $\lim_{n \rightarrow \infty} n^2 \left( \frac{1}{n^3 + 1^3} + \frac{1}{n^3 + 2^3} + \cdots + \frac{1}{n^3 + n^3} \right)$

- (c)  $\lim_{n \rightarrow \infty} \left( \frac{1^k + 2^k + \cdots + n^k}{n^{k+1}} \right), \quad k \geq 0$
- (d)  $\lim_{n \rightarrow \infty} \frac{1}{n} \sqrt[n]{(n+1)(n+2) \cdots (n+n)}$
- (e)  $\lim_{n \rightarrow \infty} \left( \sin \frac{n}{n^2 + 1^2} + \sin \frac{n}{n^2 + 2^2} + \cdots + \sin \frac{n}{n^2 + n^2} \right)$
- (f)  $\lim_{n \rightarrow \infty} \left( \frac{2^{1/n}}{n+1} + \frac{2^{2/n}}{n+\frac{1}{2}} + \cdots + \frac{2^{n/n}}{n+\frac{1}{n}} \right)$
- (g)  $\lim_{n \rightarrow \infty} \sqrt[n]{f\left(\frac{1}{n}\right) f\left(\frac{2}{n}\right) \cdots f\left(\frac{n}{n}\right)},$  where  $f$  is positive and continuous on  $[0,1]$ .
- (h)  $\lim_{n \rightarrow \infty} n \left( \frac{1^k + 2^k + \cdots + n^k}{n^{k+1}} - \frac{1}{k+1} \right), \quad k \geq 0.$
- (i)  $\lim_{n \rightarrow \infty} \left( \frac{1^k + 3^k + \cdots + (2n-1)^k}{n^{k+1}} \right), \quad k \geq 0.$
- (j)  $\lim_{n \rightarrow \infty} \left\{ \left( 1 + \frac{1}{2n} \right) \left( 1 + \frac{3}{2n} \right) \cdots \left( 1 + \frac{2n-1}{2n} \right) \right\}^{\frac{1}{2n}}.$

5. **(IMC 2024)** For  $n = 1, 2, \dots$  let

$$S_n = \log \left( \sqrt[n^2]{1^1 \cdot 2^2 \cdot \dots \cdot n^n} \right) - \log(\sqrt{n}),$$

where  $\log$  denotes the natural logarithm. Find  $\lim_{n \rightarrow \infty} S_n$ .

6. **(ISI 2024)** Find the limit:

$$\lim_{n \rightarrow \infty} \frac{2 \log 2 + 3 \log 3 + \cdots + n \log n}{n^2 \log n}$$

## 1.2. Based on basic properties

1. **(MVT I)** Let  $f$  be continuous and  $g$  be integrable on  $[a, b]$  and assume that  $g$  is positive. Show that there exists  $c \in [a, b]$  such that

$$\int_a^b f(x)g(x) \, dx = f(c) \int_a^b g(x) \, dx.$$

2. Let  $f$  be continuous and non-negative on  $[a, b]$ . If  $\int_a^b f(x) \, dx = 0$ , then show that  $f$  must be identically zero on  $[a, b]$ . Is the statement true if  $f$  is not continuous?

3. Suppose  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function satisfying

$$\int_0^1 f(x)(1 - f(x)) \, dx = \frac{1}{4}.$$

Find all such functions  $f$ .

4. Suppose  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function satisfying

$$\int_0^1 f(x)(x - f(x)) \, dx = \frac{1}{12}.$$

Find all such functions  $f$ .

5. **(SEEMOUS 2013)** Find all continuous functions  $f : [1, 8] \rightarrow \mathbb{R}$  such that

$$\int_1^2 f(t^3)^2 \, dt + 2 \int_1^2 f(t^3) \, dt = \frac{2}{3} \int_1^8 f(t) \, dt - \int_1^2 (t^2 - 1)^2 \, dt.$$

6. Find all continuous functions  $f : [0, 1] \rightarrow \mathbb{R}$  satisfying

$$\int_0^1 f(x) \, dx = \frac{1}{3} + \int_0^1 f^2(x^2) \, dx.$$

7. **(IMC 2010)** Let  $0 < a < b$ . Prove that

$$\int_a^b (1 + x^2)e^{-x^2} \, dx \geq e^{-a^2} - e^{-b^2}.$$

8. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a continuous. Define  $g(x) = \int_0^x (x - t)f(t) \, dt$  for every  $x \in \mathbb{R}$ . Show that  $g'' = f$ .

9. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be continuous and  $\alpha \neq 0$ . If

$$g(x) = \frac{1}{\alpha} \int_0^x f(t) \sin(\alpha(x - t)) \, dt,$$

then show that  $f(x) = g''(x) + \alpha^2 g(x)$ .

10. **(VJIMC 2024)** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a continuously differentiable function. Prove that

$$\left| f(1) - \int_0^1 f(x) \, dx \right| \leq \frac{1}{2} \max_{x \in [0, 1]} |f'(x)|.$$

11. Let  $f, g$  be polynomials of degree  $n$  such that

$$\int_0^1 x^k f(x) dx = \int_0^1 x^k g(x) dx$$

holds for each  $k = 0, 1, \dots, n$ . Show that  $f \equiv g$ .

12. Let  $f : [0, \infty) \rightarrow \mathbb{R}$  a non-decreasing function. Then show this inequality holds for all  $x, y, z$  such that  $0 \leq x < y < z$ :

$$(z - x) \int_y^z f(u) du \geq (z - y) \int_x^z f(u) du.$$

13. **(Putnam 1964)** Let  $a \in [0, 1]$ . Find all continuous functions  $f : [0, 1] \rightarrow [0, \infty)$  such that

$$\int_0^1 f(x) dx = 1, \quad \int_0^1 x f(x) dx = a, \quad \int_0^1 x^2 f(x) dx = a^2.$$

14. Let  $n > 1$  be an integer and let  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function such that

$$\int_0^1 f(x) dx = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}.$$

Prove that there is a real number  $x_0 \in (0, 1)$ , such that  $f(x_0) = \frac{1 - x_0^n}{1 - x_0}$ .

### 1.3. Cauchy-Schwarz and FTC and more...

1. Show that if  $f$  is integrable on  $[a, b]$ , then,

$$\left( \int_a^b f(x) \sin x dx \right)^2 + \left( \int_a^b f(x) \cos x dx \right)^2 \leq (b - a) \int_a^b f^2(x) dx$$

2. **(VJIMC 2024)** Suppose that  $f : [-1, 1] \rightarrow \mathbb{R}$  is continuous and satisfies

$$\left( \int_{-1}^1 e^x f(x) dx \right)^2 \geq \left( \int_{-1}^1 f(x) dx \right) \left( \int_{-1}^1 e^{2x} f(x) dx \right).$$

Prove that there exists a point  $c \in (-1, 1)$  such that  $f(c) = 0$ .

3. Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function such that  $\int_0^1 f(x) dx = 1$ . Show that there exists a point  $c \in (0, 1)$  such that  $f(c) = 3c^2$ .

4. Let  $f : [0, 1] \rightarrow \mathbb{R}$  be integrable such that

$$\int_0^1 x f(x) \, dx = 1.$$

Prove that  $\int_0^1 f(x)^2 \, dx \geq 3$ .

5. **(IMC 2022)** Let  $f : [0, 1] \rightarrow (0, \infty)$  be an integrable function such that  $f(x)f(1-x) = 1$  for all  $x \in [0, 1]$ . Prove that  $\int_0^1 f(x) \, dx \geq 1$ .

6. Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuously differentiable with  $f(a) = a$  and  $f(b) = b$ . Prove that

$$\int_a^b f'(x)^2 \, dx \geq b - a.$$

7. **(Putnam 2006 B5)** For each continuous function  $f : [0, 1] \rightarrow \mathbb{R}$ , let  $I(f) = \int_0^1 x^2 f(x) \, dx$  and  $J(f) = \int_0^1 x (f(x))^2 \, dx$ . Find the maximum value of  $I(f) - J(f)$  over all such functions  $f$ .

8. **(Romania District Olympiad 2006)** Let  $f_1, f_2, \dots, f_n : [0, 1] \rightarrow (0, \infty)$  be  $n$  continuous functions,  $n \geq 1$ , and let  $\sigma$  be a permutation of the set  $\{1, 2, \dots, n\}$ . Prove that

$$\prod_{i=1}^n \int_0^1 \frac{f_i^2(x)}{f_{\sigma(i)}(x)} \, dx \geq \prod_{i=1}^n \int_0^1 f_i(x) \, dx.$$

9. **(Romania District Olympiad 2003)** Let  $a$  and  $b$  be real numbers with  $a < b$ , and let  $f$  and  $g$  be continuous functions from  $[a, b]$  to  $(0, \infty)$  such that  $\int_a^b f(x) \, dx = \int_a^b g(x) \, dx$  but  $f \neq g$ . For every positive integer  $n$ , define

$$I_n = \int_a^b \frac{(f(x))^{n+1}}{(g(x))^n} \, dx.$$

Show that  $I_1, I_2, I_3, \dots$  is an increasing sequence.

#### 1.4. By Parts, Substitution and more...

1. Show that if  $f$  is continuous and even on  $[-a, a]$ ,  $a > 0$ , then,

$$\int_{-a}^a \frac{f(x)}{1 + e^x} \, dx = \int_0^a f(x) \, dx.$$

2. Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function. Prove that

$$\int_0^\pi x f(\sin x) \, dx = \pi \int_0^{\frac{\pi}{2}} f(\sin x) \, dx.$$

3. **(VJIMC 2024)** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a continuously differentiable function. Prove that

$$\left| f(1) - \int_0^1 f(x) \, dx \right| \leq \frac{1}{2} \max_{x \in [0, 1]} |f'(x)|.$$

4. **(IMC 1998)** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function satisfying  $xf(y) + yf(x) \leq 1$  for every  $x, y \in [0, 1]$ .

(a) Show that  $\int_0^1 f(x) \, dx \leq \frac{\pi}{4}$ .

- (b) Find such a function for which equality occurs.

5. Let  $f \in \mathcal{C}([a, b])$ , then find  $\lim_{n \rightarrow \infty} \int_a^b f(x) \cos(nx) \, dx$ .

6. Let  $f \in \mathcal{C}^2([0, 2\pi])$  and  $f''(x) \geq 0$  for all  $x \in [0, 2\pi]$ . Show that

$$\int_0^{2\pi} f(x) \cos x \, dx \geq 0.$$

7. Define  $I_n = \int_0^1 \frac{x^n}{\sqrt{1+x^2}} \, dx$  for every  $n \in \mathbb{N}$ . Prove that  $\lim_{n \rightarrow \infty} nI_n = \frac{1}{\sqrt{2}}$ .

8. **(IMC 1996)** Evaluate the definite integral

$$\int_{-\pi}^{\pi} \frac{\sin nx}{(1+2^x) \sin x} \, dx,$$

where  $n$  is a natural number.

9. **(CMI 2019)** Evaluate  $\int_0^\infty (1+x^2)^{-(m+1)} \, dx$  where  $m \in \mathbb{N}$ .
10. **(ISI 2016)** Suppose  $f$  is a differentiable function such that  $f(f(x)) = x$  where  $x \in [0, 1]$ . Also  $f(0) = 1$ . Find the value of

$$\int_0^1 (x - f(x))^{2016} \, dx.$$

11. Prove the following equality:

$$\int_0^{\frac{\pi}{2}} \frac{\sin x \cos x}{(x+1)} \, dx = \frac{1}{2} \left( \frac{1}{2} + \frac{1}{\pi+2} - \int_0^\pi \frac{\cos x}{(x+2)^2} \, dx \right).$$

## 1.5. Miscellaneous problems

1. (**Romania District Olympiad 2001**) Consider a continuous function  $f : [0, 1] \rightarrow \mathbb{R}$  such that for any third degree polynomial function  $P : [0, 1] \rightarrow [0, 1]$ , we have

$$\int_0^1 f(P(x)) dx = 0.$$

Prove that  $f(x) = 0 \forall x \in [0, 1]$ .

2. Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuously differentiable with  $f(a) = f(b) = 0$  and  $\int_a^b f(x)^2 dx = 1$ . Show that

$$\int_a^b (f'(x))^2 dx \cdot \int_a^b (xf(x))^2 dx \geq \frac{1}{4}.$$

3. (**Romania MO 2002**) Let  $f : [0, \infty) \rightarrow \mathbb{R}$  be a continuous and bounded function such that

$$x \int_x^{x+1} f(t) dt = \int_0^x f(t) dt$$

for any  $x \in [0, \infty)$ . Prove that  $f(x)$  is a constant function.

4. Let  $P$  be a polynomial of degree  $n$  such that

$$\int_0^1 x^k P(x) dx = 0 \quad \text{for } k = 1, 2, \dots, n.$$

Prove that

$$\int_0^1 (P(x))^2 dx = (n+1)^2 \left( \int_0^1 P(x) dx \right)^2.$$

5. Let  $q$  be positive and decreasing on  $[a, b]$ ,  $a > 0$ . Show that

$$\frac{\int_a^b xq^2(x) dx}{\int_a^b xq(x) dx} \leq \frac{\int_a^b q^2(x) dx}{\int_a^b q(x) dx}.$$

6. Show that if  $f$  is continuous on  $[0, 1]$ , differentiable on  $(0, 1)$ ,  $f(0) = 0$ , and  $0 < f'(x) \leq 1$  on  $(0, 1)$ , then

$$\left( \int_0^1 f(x) dx \right)^2 \geq \int_0^1 (f(x))^3 dx.$$

Show also that equality holds if and only if  $f(x) = x$ .

7. Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuously differentiable with  $f(a) = 0$ . Show that

$$\int_a^b f^2(x) \, dx \leq \frac{(b-a)^2}{2} \int_a^b (f'(x))^2 \, dx.$$

8. Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuously differentiable with  $f(a) = f(b) = 0$ . Show that

$$\int_a^b f^2(x) \, dx \leq \frac{(b-a)^2}{8} \int_a^b (f'(x))^2 \, dx.$$

9. (**Putnam 2007**) Suppose that  $f : [0, 1] \rightarrow \mathbb{R}$  has a continuous derivative and that  $\int_0^1 f(x) \, dx = 0$ . Prove that for every  $\alpha \in (0, 1)$ ,

$$\left| \int_0^\alpha f(x) \, dx \right| \leq \frac{1}{8} \max_{0 \leq x \leq 1} |f'(x)|$$

10. (**Putnam 2008 A4**) Define  $f : \mathbb{R} \rightarrow \mathbb{R}$  by

$$f(x) = \begin{cases} x & \text{if } x \leq e \\ xf(\ln x) & \text{if } x > e \end{cases}$$

Does  $\sum_{n=1}^{\infty} \frac{1}{f(n)}$  converge?

11. (**Romania MO 2008**) For continuously differentiable function  $f : [0, 1] \rightarrow \mathbb{R}$  with  $f(1/2) = 0$ , show that

$$\left( \int_0^1 f(x) \, dx \right)^2 \leq \frac{1}{12} \int_0^1 (f'(x))^2 \, dx.$$

12. (**Modified version of IMC 1995 P2**) Let  $f$  be a continuous function on  $[0, 1]$  such that for every  $x \in [0, 1]$ , we have

$$\int_x^1 f(t) \, dt \geq \frac{1-x^2}{2}.$$

Show that

$$\int_0^1 f(t)^2 \, dt \geq \int_0^1 xf(x) \, dx.$$

13. **(Romanian District Olympiad 2025)** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function. Define  $\tilde{f} : [0, 1] \rightarrow \mathbb{R}$  as

$$\tilde{f}(x) = \begin{cases} \frac{1}{x} \cdot \int_0^x f(t) \, dt & \text{for } x > 0; \\ f(0) & \text{for } x = 0. \end{cases}$$

Show that:

- (a)  $\tilde{f}$  is continuous in 0 and differentiable on  $(0, 1]$ .  
(b) the following equality takes place:

$$\int_0^1 f^2(x) \, dx = \left( \int_0^1 f(x) \, dx \right)^2 + \int_0^1 \left( f(x) - \tilde{f}(x) \right)^2 \, dx.$$

14. **(Romania MO 2025)** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a differentiable function such that its derivative is an integrable function on  $[0, 1]$ , and  $f(1) = 0$ . Prove that

$$\int_0^1 (x f'(x))^2 \, dx \geq 12 \cdot \left( \int_0^1 x f(x) \, dx \right)^2$$

15. **(Romania MO 2004)** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be an integrable function such that

$$\int_0^1 f(x) \, dx = \int_0^1 x f(x) \, dx = 1.$$

Prove that  $\int_0^1 f(x)^2 \, dx \geq 4$ .

16. **(AMM 2006)** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be an integrable function such that

$$\int_0^1 f(x) \, dx = \int_0^1 x f(x) \, dx = \int_0^1 x^2 f(x) \, dx = 1.$$

Show that  $\int_0^1 f^2(x) \, dx \geq 9$ .

17. Let  $f : [0, 1] \rightarrow [0, \infty)$  be a continuous function such that

$$f(t)^2 \leq 1 + 2 \int_0^t f(s) \, ds, \quad \forall t \in [0, 1].$$

Prove that  $f(t) \leq 1 + t$ , for all  $t \in [0, 1]$ .

18. Let  $f : [1, \infty) \rightarrow \mathbb{R}$  be such that  $f(1) = 1$  and

$$f'(x) = \frac{1}{x^2 + f(x)^2}, \quad \forall x \geq 1.$$

Prove that  $\lim_{x \rightarrow \infty} f(x)$  exists and is less than  $1 + \frac{\pi}{4}$ .

19. Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function satisfying  $f(x) + f(1 - x) = 1$  for every  $x \in [0, 1]$ . For every  $n \in \mathbb{N}$ , let  $f^n$  denote  $f$  composed  $n$  times. Evaluate  $\int_0^1 f^{2026}(x) dx$ .

20. Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function. Prove that

(a)  $\lim_{n \rightarrow \infty} \int_0^1 x^n f(x) dx = 0,$

(b)  $\lim_{n \rightarrow \infty} n \int_0^1 x^n f(x) dx = f(1).$

21. Suppose  $f : [0, \infty) \rightarrow \mathbb{R}$  is continuous. Define  $a_n = \int_0^1 f(x + n) dx$ , for every  $n \geq 0$ . Suppose also that  $\lim_{n \rightarrow \infty} a_n = a$ . Find the limit

$$\lim_{n \rightarrow \infty} \int_0^1 f(nx) dx.$$

22. Let  $f$  be a non-negative integrable function on  $[0, 1]$  such that  $\int_0^1 f(x) dx = 1$ .  
Prove that

$$\int_0^1 \left( x - \int_0^1 u f(u) du \right)^2 f(x) dx \leq \frac{1}{4}.$$

23. Let  $a, b, L > 0$  and  $f \in \mathcal{C}([0, L])$  be such that  $f(0) = 0$  and, for  $0 \leq x \leq L$ ,

$$f'(x)^2 \leq a^2 + 2b^2 \int_0^x |f(u)f'(u)| du.$$

Prove that for any  $0 \leq x \leq L$ ,

$$|f(x)| \leq \frac{a}{b}(e^{bx} - 1)$$

and hence show that

$$|f'(x)| \leq ae^{bx}.$$

24. Prove that the equation

$$f(x) \int_a^x g(t) dt = g(x) \int_x^b f(t) dt$$

has at least one solution in the interval  $(a, b)$ .

25. **(Romania District Olympiad 2016)** Let  $f : \mathbb{R} \rightarrow (0, \infty)$  be a continuous and periodic function having a period of 2, and such that the integral  $\int_0^2 \frac{f(1+x)}{f(x)} dx$  exists. Show that

$$\int_0^2 \frac{f(1+x)}{f(x)} dx \geq 2,$$

with equality if and only if 1 is also a period of  $f$ .

26. **(Romania MO 2009)** Find all functions  $f \in \mathcal{C}^1([0, 1])$  that satisfy  $f(1) = -\frac{1}{6}$  and

$$\int_0^1 (f'(x))^2 dx \leq 2 \int_0^1 f(x) dx.$$

27. **(Mathematical Magazine 2011)** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuously differentiable function such that  $f(0) = f(1) = -\frac{1}{6}$ . Prove that

$$\int_0^1 (f'(x))^2 dx \geq 2 \int_0^1 f(x) dx + \frac{1}{4}.$$

28. **(Romania MO 2012)** Let  $f : [0, \infty) \rightarrow \mathbb{R}$  be a continuous function such that

$$\int_0^n f(x)f(n-x) dx = \int_0^n f^2(x) dx,$$

for any natural number  $n \geq 1$ . Prove that  $f$  is a periodic function.

29. **(Romania MO 2013)** Determine continuous functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$(a^2 + ab + b^2) \int_a^b f(x) dx = 3 \int_a^b x^2 f(x) dx, \quad \forall a, b \in \mathbb{R}.$$

30. **(Romania MO 2007)** Let  $\mathcal{F}$  be the set of functions  $f : [0, 1] \rightarrow \mathbb{R}$  that are differentiable, with continuous derivative, and  $f(0) = 0$ ,  $f(1) = 1$ . Find the minimum of  $\int_0^1 \sqrt{1+x^2} \cdot (f'(x))^2 dx$  (where  $f \in \mathcal{F}$ ) and find all functions  $f \in \mathcal{F}$  for which this minimum is attained.

31. **(Romania District Olympiad 2002)** Let  $f$  be a continuous and periodic function  $f : \mathbb{R} \rightarrow [0, \infty)$  of period 1. Show:

(a)  $a \in \mathbb{R} \implies \int_a^{a+1} f(x) dx = \int_0^1 f(x) dx.$

(b)  $\lim_{n \rightarrow \infty} \int_0^1 f(x)f(nx) dx = \left( \int_0^1 f(x) dx \right)^2.$

32. **(IMC 1998)** Let  $f(x) = 2x(1-x)$ ,  $x \in \mathbb{R}$  and denote  $f_n = f \circ f \circ \dots \circ f$ ,  $n$  times.

(a) Find  $\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx.$

(b) Now compute  $\int_0^1 f_n(x) dx.$

33. **(Romania District Olympiad 2024)** Let  $f : [0, \infty) \rightarrow \mathbb{R}$  be a differentiable function, with a continuous derivative. Given that  $f(0) = 0$  and  $0 \leq f'(x) \leq 1$  for every  $x > 0$  prove that

$$\frac{1}{n+1} \int_0^a f(t)^{2n+1} dt \leq \left( \int_0^a f(t)^n dt \right)^2,$$

for any positive integer  $n$  and real number  $a > 0$ .

34. **(Romania District Olympiad 2017)** Let  $f, g : [0, 1] \rightarrow \mathbb{R}$  be two continuous functions such that  $f(x)g(x) \geq 4x^2$ , for all  $x \in [0, 1]$ . Prove that

$$\left| \int_0^1 f(x) dx \right| \geq 1 \text{ or } \left| \int_0^1 g(x) dx \right| \geq 1.$$

35. **(SEEMOUS 2011)** For a given integer  $n \geq 1$ , let  $f : [0, 1] \rightarrow \mathbb{R}$  be a non-decreasing function. Prove that

$$\int_0^1 f(x) dx \leq (n+1) \int_0^1 x^n f(x) dx.$$

Find all non-decreasing continuous functions for which equality holds.

36. **(IberoAmerica 1998)** Suppose  $f, g$  be two continuous real function such that  $\int_0^1 f(x)^2 dx = \int_0^1 g(x)^2 dx = 1$ . Prove that there exists  $c \in [0, 1]$  such that  $f(c) + g(c) \leq 2$ .

37. **(VJIMC 2009)** Let  $E$  be the set of all continuously differentiable real valued

functions  $f$  on  $[0, 1]$  such that  $f(0) = 0$  and  $f(1) = 1$ . Define

$$J(f) = \int_0^1 (1 + x^2) f'(x)^2 dx.$$

a) Show that  $J$  achieves its minimum value at some element of  $E$ .

b) Calculate  $\min_{f \in E} J(f)$ .

38. **(VJIMC 2008)** Find all continuously differentiable functions  $f : [0, 1] \rightarrow (0, \infty)$  such that  $\frac{f(1)}{f(0)} = e$  and

$$\int_0^1 \frac{dx}{f(x)^2} + \int_0^1 f'(x)^2 dx \leq 2.$$

39. **(Romania MO 2003)** Let  $n$  be an odd integer greater than 1. Determine all continuous functions  $f : [0, 1] \rightarrow \mathbb{R}$  such that

$$\int_0^1 \left( f \left( x^{\frac{1}{k}} \right) \right)^{n-k} dx = \frac{k}{n}, \quad \text{for } k = 1, 2, \dots, n-1.$$

40. **(Putnam 2008)** Let  $F_0 = \ln x$ . For  $n \geq 0$  and  $x > 0$ , let  $F_{n+1}(x) = \int_0^x F_n(t) dt$ .

Evaluate  $\lim_{n \rightarrow \infty} \frac{n! F_n(1)}{\ln n}$ .

41. **(Putnam 1967)** If  $f$  and  $g$  are continuous and periodic functions with period 1 on the real line, then

$$\lim_{n \rightarrow \infty} \int_0^1 f(x) g(nx) dx = \left( \int_0^1 f(x) dx \right) \left( \int_0^1 g(x) dx \right).$$

42. Show that if  $f$  is continuously differentiable on  $[0, 1]$ , then,

$$\lim_{n \rightarrow \infty} n \left( \frac{1}{n} \sum_{i=1}^n f \left( \frac{i}{n} \right) - \int_0^1 f(x) dx \right) = \frac{f(1) - f(0)}{2}.$$

Using this result, calculate

$$\lim_{n \rightarrow \infty} n \left( \frac{1^k + 2^k + \dots + n^k}{n^{k+1}} - \frac{1}{k+1} \right), \quad k \geq 0.$$

43. Suppose that  $f$  is twice differentiable on  $[0, 1]$  and  $f''$  is bounded and integrable.

Show that

$$\lim_{n \rightarrow \infty} n^2 \left( \int_0^1 f(x) dx - \frac{1}{n} \sum_{i=1}^n f\left(\frac{2i-1}{2n}\right) \right) = \frac{f'(1) - f'(0)}{24}.$$

44. **(Chebyshev's Inequality)** Let  $f$  and  $g$  be two increasing functions on  $\mathbb{R}$ . Then for any reals  $a < b$ , prove that

$$(b-a) \int_a^b f(x)g(x) dx \geq \left( \int_a^b f(x) dx \right) \left( \int_a^b g(x) dx \right).$$

45. Let  $f : [0, \infty) \rightarrow [0, \infty)$  be a continuous, strictly increasing function with  $f(0) = 0$ . Prove that

$$\int_0^a f(x) dx + \int_0^b f^{-1}(x) dx \geq ab,$$

for all positive  $a, b$ , with equality if and only if  $b = f(a)$ .

46. **(Romania MO 1971)** Find all continuous functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  satisfying the equation

$$f(x) = \lambda(1+x^2) \left[ 1 + \int_0^x \frac{f(t)}{1+t^2} dt \right]$$

for all  $x \in \mathbb{R}$ . Here  $\lambda$  is a fixed real number.

47. **(Romania District Olympiad 2008)** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function such that

$$\int_0^1 f(x) dx = \int_0^1 x f(x) dx.$$

Show that there is a  $c \in (0, 1)$  such that  $f(c) = \int_0^c f(x) dx$ .

48. **(Romania District Olympiad 2007)** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function and  $g : [0, 1] \rightarrow (0, \infty)$ . Prove that if  $f$  is increasing, then

$$\int_0^t f(x)g(x) dx \cdot \int_0^1 g(x) dx \leq \int_0^t g(x) dx \cdot \int_0^1 f(x)g(x) dx.$$

49. **(Jozsef Wildt IMC 2009)** If the function  $f : [0, 1] \rightarrow (0, +\infty)$  is increasing and continuous, then for every  $a \geq 0$  the following inequality holds:

$$\int_0^1 \frac{x^{a+1}}{f(x)} dx \leq \frac{a+1}{a+2} \int_0^1 \frac{x^a}{f(x)} dx.$$

50. **(VJIMC 2022)** Determine whether there exists a differentiable function  $f : [0, 1] \rightarrow \mathbb{R}$  such that

$$f(0) = f(1) = 1, \quad |f'(x)| \leq 2 \text{ for all } x \in [0, 1] \quad \text{and} \quad \left| \int_0^1 f(x) \, dx \right| \leq \frac{1}{2}.$$

51. **(Romania MO SL 2005)** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a differentiable function with continuous derivative such that

$$\int_0^1 f(x) \, dx = \int_0^1 x f(x) \, dx = 1.$$

Show that

$$\int_0^1 (f'(x))^2 \, dx \geq 30.$$

52. Let  $f : [0, 1] \rightarrow \mathbb{R}$  be continuously differentiable on  $[0, 1]$  with  $f(1) = 0$ . Show that

$$\int_0^1 f(x)^2 \, dx \leq 4 \int_0^1 x^2 (f'(x))^2 \, dx.$$

53. **(College Mathematics Journal 2012)** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be continuously differentiable on  $[0, 1]$  with  $f(1) = 0$ . Show that

$$4 \int_0^1 x^2 (f'(x))^2 \, dx \geq \int_0^1 f(x)^2 \, dx + \left( \int_0^1 f(x) \, dx \right)^2.$$

54. **(Romania MO 1997)** Show that for any continuous function  $f : [-1, 1] \rightarrow \mathbb{R}$  we have the following inequalities:

$$\int_{-1}^1 f^2(x) \, dx \geq \frac{1}{2} \left( \int_{-1}^1 f(x) \, dx \right)^2 + \frac{3}{2} \left( \int_{-1}^1 x f(x) \, dx \right)^2.$$

55. **(SEEMOUS 2008)** Let  $n$  be a positive integer and  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function such that

$$\int_0^1 x^k f(x) \, dx = 1$$

for every  $k \in \{0, 1, \dots, n-1\}$ . Prove that

$$\int_0^1 f(x)^2 \, dx \geq n^2.$$

56. **(Madhava 2026)** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function with  $\int_0^1 x(x-1)^2 f(x) dx = 0$ . Prove that there is a real number  $c \in (0, 1)$  such that

$$\int_0^c x^2 f(x) dx = c \int_0^c x f(x) dx.$$

57. **(Madhava 2026)** Let  $a_1, a_2, \dots, a_n$  be real numbers. Prove that

$$\sum_{1 \leq i, j \leq n} \frac{a_i a_j}{i+j-1} \geq 0.$$

58. Let  $\mathcal{F}$  be the set of all continuously differentiable functions satisfying  $f(0) = 1$  and  $f(1) = e$ . Find

$$\inf_{f \in \mathcal{F}} \left( \int_0^1 e^x \left[ \frac{d}{dx} (\log f(x)) \cdot f(x) \right]^2 dx \right).$$

59. Let  $f : [0, 1] \rightarrow (0, \infty)$  be a continuously differentiable functions satisfying

$$\int_0^1 f(x) dx = 1.$$

Prove that

$$\left( \int_0^1 f(x) \left[ \frac{d}{dx} \log f(x) \right]^2 dx \right) \left( \int_0^1 x^2 f(x) dx \right) \geq (f(1) - 1)^2.$$

60. Let  $f : [0, 1] \rightarrow [0, \infty)$  be a continuously differentiable function such that  $f(0) = 1$  and  $f(1) = 0$ . Prove that

$$\left( \int_0^1 \frac{f'(x)^2}{f(x)^{4/3}} dx \right) \left( \int_0^1 x^2 f(x)^{2/3} dx \right) \geq \frac{9}{4} \left( \int_0^1 f(x)^{2/3} dx \right).$$

61. **(Jozsef Wildt IMC 2019)** If  $f : [a, b] \rightarrow (0, \infty)$ ,  $0 < a \leq b$ , with  $f$  continuously differentiable, prove that

$$\int_a^b \frac{f'(x) \sqrt{f(x)}}{f^3(x) + 1} dx \leq \tan^{-1} \left( \frac{f(b) - f(a)}{1 + f(a)f(b)} \right).$$

62. If  $f : [0, 1] \rightarrow \mathbb{R}$  is differentiable, then prove that  $\exists c \in (0, 1)$  such that

$$\int_0^1 f(t) \, dt = f(0) + \frac{1}{2}f'(c).$$

63. Let  $f : \left[0, \frac{\pi}{4}\right] \rightarrow \mathbb{R}$  be continuous. Prove that  $\exists c \in \left(0, \frac{\pi}{4}\right)$  such that

$$2 \cos 2c \int_0^{\pi/4} f(t) \, dt = f(c).$$

64. Let  $f : [0, a] \rightarrow \mathbb{R}$  be such that  $f''(x) \geq 0$  for all  $x \in [0, a]$ . Prove that

$$\int_0^a f(t) \, dt \geq af\left(\frac{a}{2}\right).$$